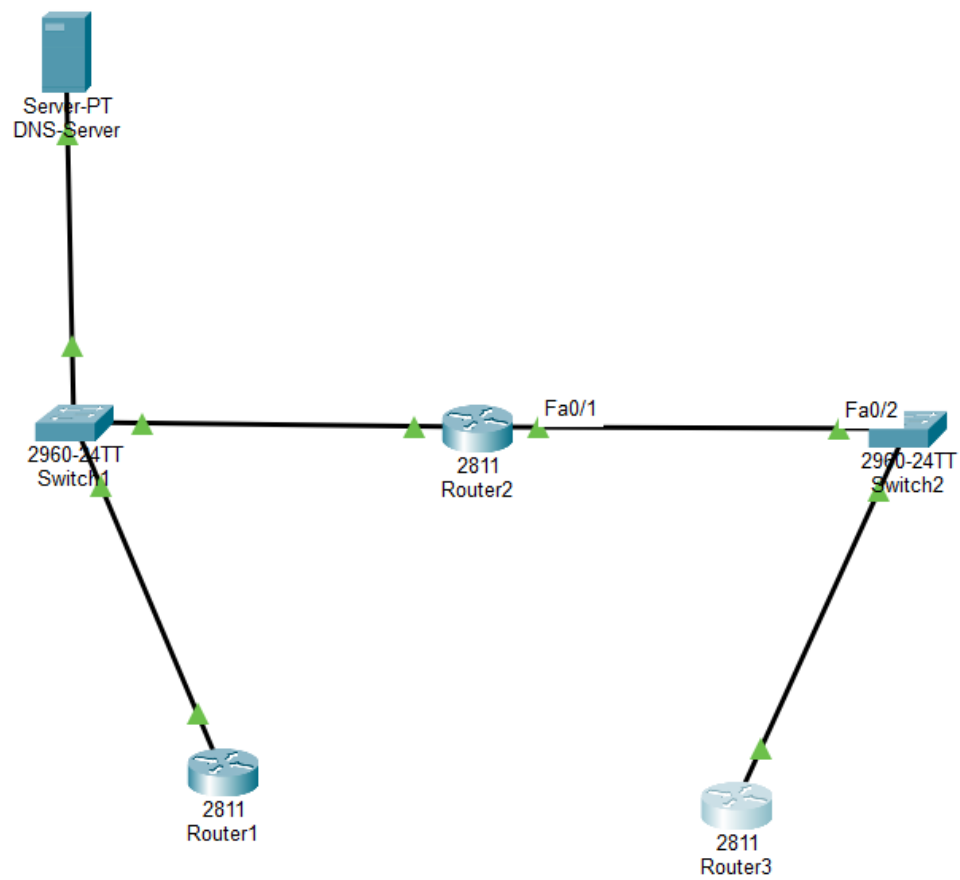


Week 3: Life of a Packet (10 Marks)

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Date: 14 August 2026



16)R1

 Router1

Physical Config CLI Attributes

IOS Command Line Interface

```

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNIL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#ip address 10.10.10.1 255.0.0.0
Router(config-if)#
Router(config-if)#enable
      ^
% Invalid input detected at '^' marker.

Router(config-if)#enable
      ^
% Invalid input detected at '^' marker.

Router(config-if)#
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNIL/Z.
Router(config)#ip route 10.10.20.2 255.255.255.0 10.10.10.0
%Inconsistent address and mask
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNIL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
    
```

R2)

 Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(Config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#

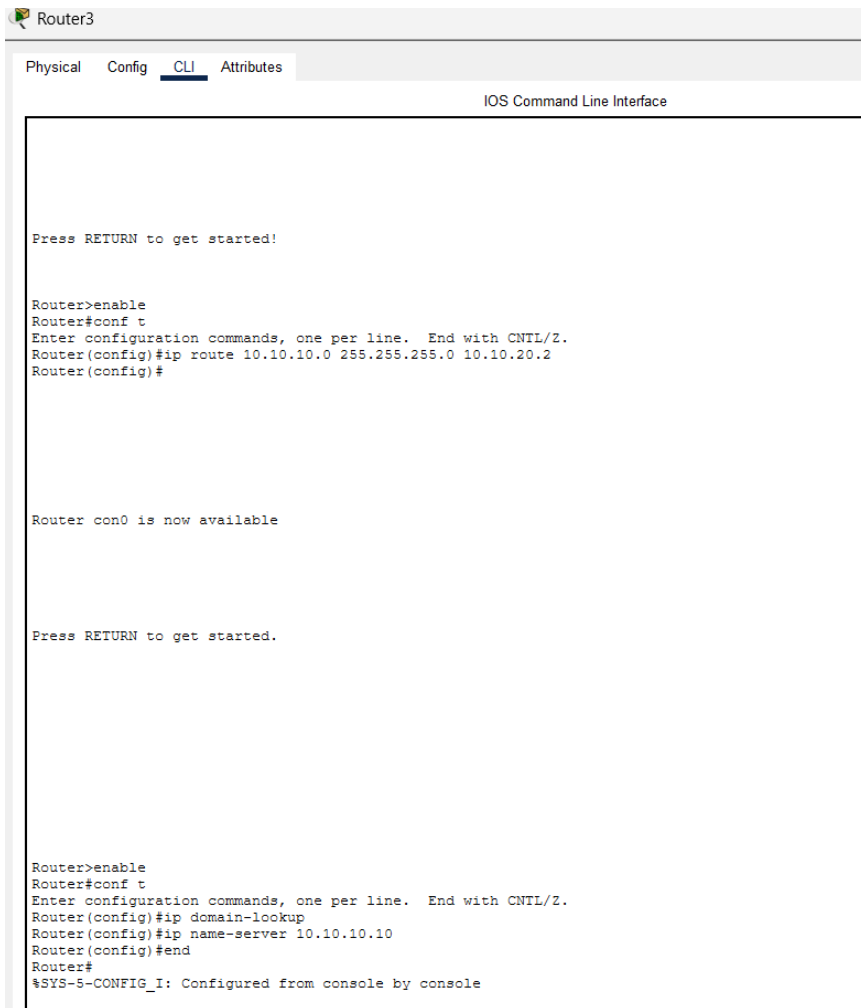
Router con0 is now available

Press RETURN to get started.

Press RETURN to get started!

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

R3)



17)

```
Router>enable
Router#ping 10.10
Router#end
Translating "end"...domain server (10.10.10.10)
% Unknown command or computer name, or unable to find computer address

Router#enable
Router#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
..!!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/1 ms

Router#|
```

18)

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

Router>enable
Router#ping 10.10
Router#end
Translating "end"...domain server (10.10.10.10)
% Unknown command or computer name, or unable to find computer address

Router#enable
Router#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
..!!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/1 ms

Router#clear ip arp
^
% Invalid input detected at '^' marker.

Router#clear arp-cache
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 00D0.FFBE.1B01 ARPA FastEthernet0/0
Internet 10.10.10.10 0 00E0.8FBC.ED85 ARPA FastEthernet0/0
Internet 10.10.20.1 0 0090.2B24.1901 ARPA FastEthernet0/0
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 00D0.FFBE.1B01 ARPA FastEthernet0/0
Internet 10.10.10.2 0 0090.2B24.1901 ARPA FastEthernet0/0
Internet 10.10.10.10 2 00E0.8FBC.ED85 ARPA FastEthernet0/0
Internet 10.10.20.1 2 0090.2B24.1901 ARPA FastEthernet0/0
Router#
```

19)

```
Router#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 10.10.10.1          -    00D0.FFBE.1B01  ARPA   FastEthernet0/0
Internet 10.10.10.2          0    0090.2B24.1901  ARPA   FastEthernet0/0
Internet 10.10.10.10         2    00E0.8FBC.ED85  ARPA   FastEthernet0/0
Internet 10.10.20.1          2    0090.2B24.1901  ARPA   FastEthernet0/0
R1 Router#
```

R2)

```
Press RETURN to get started!

Router>enable
Router#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 10.10.10.1        17    00D0.FFBE.1B01  ARPA   FastEthernet0/0
Internet 10.10.10.2          -    0090.2B24.1901  ARPA   FastEthernet0/0
Internet 10.10.20.1        17    0090.0C26.9B01  ARPA   FastEthernet0/1
Internet 10.10.20.2          -    0090.2B24.1902  ARPA   FastEthernet0/1
Router#
```

R3)

```
Press RETURN to get started.

Router>enable
Router#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 10.10.20.1          -    0090.0C26.9B01  ARPA   FastEthernet0/0
Internet 10.10.20.2        18    0090.2B24.1902  ARPA   FastEthernet0/0
Router#
```

20) A routing table entry only tells you the next-hop IP address and which interface to send the packet out. An ARP entry gives you the MAC address of that next-hop IP address. The router needs the MAC to build the layer 2 Ethernet frame to send the packet on the local network.

21) Because IP routing and Ethernet delivery are 2 different jobs. The routing table tells the router where to send the packet logically- the next-hop IP and exit interface.